User Scenarios List

short description of all the user scenarios

Nr	Name	Description
US_01	View borrowing history	Member: A logged-in member views the list of books they have previously borrowed.
US_02	View overdue books	Member: A logged-in member checks which borrowed books are overdue.
US_03	View fine amount	Member: A logged-in member sees the total fine amount for overdue books.
US_04	Receive overdue notifications	Member: A member receives notifications when a borrowed book becomes overdue.
US_05	View fine status	Member: A logged-in member sees whether their fines are paid or unpaid.
US_06	View members with fines	Staff: Staff views a list of members who have unpaid or overdue fines.
US_07	Adjust member fine	Administrator: Admin resets or adjusts a member's fine in special cases.
US_08	Mark fine as paid	Staff: Staff marks a member's fine as paid.
US_09	View payment history	Member: A logged-in member views the history of their paid fines.
US_10	Generate fine reports	Administrator: Admin generates reports about collected fines and overdue penalties.

User Scenarios Extended

US_01: View borrowing history

1. Member logs into the Librario platform.
2. Member navigates to the borrowing history page.
3. System retrieves the list of books previously borrowed by the member.
4. System displays book details such as title, borrow date, return date, and status.
5. If the member has no borrowing history: System shows a message like **"No borrowing history found."**

US_02: View overdue books

1. Member logs into the Librario platform.
2. Member opens the borrowed books or overdue books section.
3. System checks all currently borrowed books.
4. System identifies books whose return date has passed.
5. System displays the overdue books list.
6. If there are no overdue books: System shows **“No overdue books.”**

US_03: View fine amount

1. Member logs into the Librario platform.
2. Member navigates to the fines section.
3. System calculates or retrieves the total fine amount for overdue books.
4. System displays the amount owed.
5. If the member has no fines: System shows **“No fines due.”**

US_04: Receive overdue notifications

1. A borrowed book passes its due date.
2. System detects that the book is overdue.
3. System generates a notification for the member.
4. Member receives the notification in the system or by email.
5. Notification includes the overdue book and related fine information.
6. If notification delivery fails: System logs the failure and may try again later.

US_05: View fine status

1. Member logs into the Librario platform.
2. Member opens the fines or payments section.
3. System retrieves all fines linked to the member account.
4. System displays whether each fine is **paid** or **unpaid**.
5. If no fines exist: System shows **“No fines available.”**

US_06: View members with fines

1. Librarian logs into the Librario platform.
2. Librarian navigates to the fines management section.
3. System retrieves members with unpaid or overdue fines.
4. System displays the list with member name, fine amount, and fine status.
5. Librarian reviews the list to notify members or restrict borrowing if needed.
6. If no members have fines: System shows **“No members with unpaid or overdue fines.”**

US_07: Adjust member fine

1. Administrator logs into the system.
2. Administrator accesses the fine management section.
3. Administrator searches for a member.
4. System displays the member’s fine details.
5. Administrator updates or resets the fine amount.
6. System saves the changes and confirms the update.

7. If member not found or invalid value entered: Show error message.

US_08: Mark fine as paid

1. Librarian logs into the system.
2. Librarian navigates to fines management.
3. Librarian selects a specific fine.
4. System displays the fine details.
5. Librarian marks the fine as **paid**.
6. System updates the status and confirms the action.
7. If already paid: Show **"Fine already marked as paid."**

US_09: View payment history

1. Member logs into the Librario platform.
2. Member navigates to the payment history section.
3. System retrieves all paid fine records for that member.
4. System displays the payment history with details such as amount, payment date, and related book.
5. If there are no paid fines: System shows **"No payment history found."**

US_10: Generate fine reports

1. Administrator logs into the system.
2. Administrator navigates to the reports section.
3. Administrator selects a report type (fines, overdue penalties).
4. System collects the required data.
5. System generates and displays the report.
6. Administrator can download or review the report.
7. If no data is available: Show **"No report data available."**

Use Cases

UC_01: View borrowing history

UC Name	UC_01 — View borrowing history
Dependency	None
Actors	Primary actor: Member
Preconditions	Member is logged into the system.
Description of the Main Sequence	1. Member navigates to the borrowing history page. 2. System retrieves all previously borrowed books. 3. System displays the list with details (title, borrow date, return date, status).
Description of the Alternative Sequence	• No borrowing history: System displays **"No borrowing history found."** • Data retrieval error: System shows an error message.

Non-functional requirements	Page must load within 2 seconds. Data must be displayed clearly and organized.
Postconditions	Member successfully views their borrowing history.

UC_02: View overdue books

UC Name	UC_02 — View overdue books
Dependency	None
Actors	Primary actor: Member
Preconditions	Member is logged into the system.
Description of the Main Sequence	1. Member navigates to the borrowed books or overdue books section. 2. System checks all currently borrowed books. 3. System identifies books whose return date has passed. 4. System displays the overdue books list with relevant details.
Description of the Alternative Sequence	• No overdue books: System displays **"No overdue books."** • Data retrieval error: System shows an error message.
Non-functional requirements	Page must load within 2 seconds. Data must be displayed clearly and accurately.
Postconditions	Member successfully views all overdue books.

UC_03: View fine amount

UC Name	UC_03 — View fine amount
Dependency	None
Actors	Primary actor: Member
Preconditions	Member is logged into the system.
Description of the Main Sequence	1. Member navigates to the fines section. 2. System calculates or retrieves the total fine amount for overdue books. 3. System displays the amount owed.
Description of the Alternative Sequence	• No fines: System displays **"No fines due."** • Data retrieval error: System shows an error message.
Non-functional requirements	Fine information must load within 2 seconds. Amount shown must be accurate and up to date.
Postconditions	Member successfully views the total fine amount.

UC_04: Receive overdue notifications

UC Name	UC_04 — Receive overdue notifications
Dependency	None
Actors	Primary actor: Member Secondary actor: System
Preconditions	Member has borrowed a book. The due date has passed.
Description of the Main Sequence	1. A borrowed book passes its due date. 2. System detects that the book is overdue. 3. System generates a notification for the member. 4. Member receives the notification in the system or by email. 5. Notification includes the overdue book and related fine information.
Description of the Alternative Sequence	• Notification delivery fails: System logs the failure and may try again later.
Non-functional requirements	Notification must be generated promptly after the due date passes. Notification content must be clear and accurate.
Postconditions	Member is informed that the book is overdue and may have a related fine.

UC_05: View fine status

UC Name	UC_05 — View fine status
Dependency	None
Actors	Primary actor: Member
Preconditions	Member is logged into the system.
Description of the Main Sequence	1. Member navigates to the fines or payments section. 2. System retrieves all fines linked to the member account. 3. System displays each fine with its status (paid or unpaid).
Description of the Alternative Sequence	• No fines exist: System displays **"No fines available."** • Data retrieval error: System shows an error message.
Non-functional requirements	Fine status must load within 2 seconds. Information must be accurate and clearly presented.
Postconditions	Member successfully views the status of all fines.

UC_06: View members with fines

UC Name	UC_06 — View members with fines
Dependency	None
Actors	Primary actor: Librarian(staff)
Preconditions	Librarian is logged into the system.
Description of the Main Sequence	1. Librarian navigates to the fines management section. 2. System retrieves members with unpaid or overdue fines. 3. System displays the list with member name, fine amount, and fine status. 4. Librarian reviews the list to notify members or restrict borrowing if needed.
Description of the Alternative Sequence	• No members have fines: System displays **"No members with unpaid or overdue fines."** • Data retrieval error: System shows an error message.
Non-functional requirements	The list must load within 2 seconds. Data must be accurate and clearly organized.
Postconditions	Librarian successfully views members with unpaid or overdue fines.

UC_07: Adjust member fine

UC Name	UC_07 — Adjust member fines
Dependency	UC_06 — View members with fines
Actors	Primary actor: Administrator
Preconditions	Administrator is logged into the system. Members with fines exist in the system.
Description of the Main Sequence	1. Administrator navigates to the fine management section. 2. Administrator searches for a member. 3. System displays the member's fine details. 4. Administrator updates or resets the fine amount. 5. System saves the changes and confirms the update.
Description of the Alternative Sequence	• Member not found: System displays an error message. • Invalid fine value entered: System shows an error message.
Non-functional requirements	Updates must be applied immediately. Changes must be accurate and reliably stored.
Postconditions	Member's fine is updated successfully in the system.

UC_08: Mark fine as paid

UC Name	UC_08 — Mark fine as paid
Dependency	UC_06 — View members with fines
Actors	Primary actor: Librarian(staff)
Preconditions	Librarian is logged into the system. An unpaid fine exists for a member.
Description of the Main Sequence	1. Librarian navigates to fines management. 2. Librarian selects a specific fine. 3. System displays the fine details. 4. Librarian marks the fine as paid. 5. System updates the status and confirms the action.
Description of the Alternative Sequence	• Fine already paid: System displays **"Fine already marked as paid."** • Fine record not found: System shows an error message.
Non-functional requirements	Status update must be applied immediately. Payment status must be stored accurately.
Postconditions	Fine status is updated to **paid** in the system.

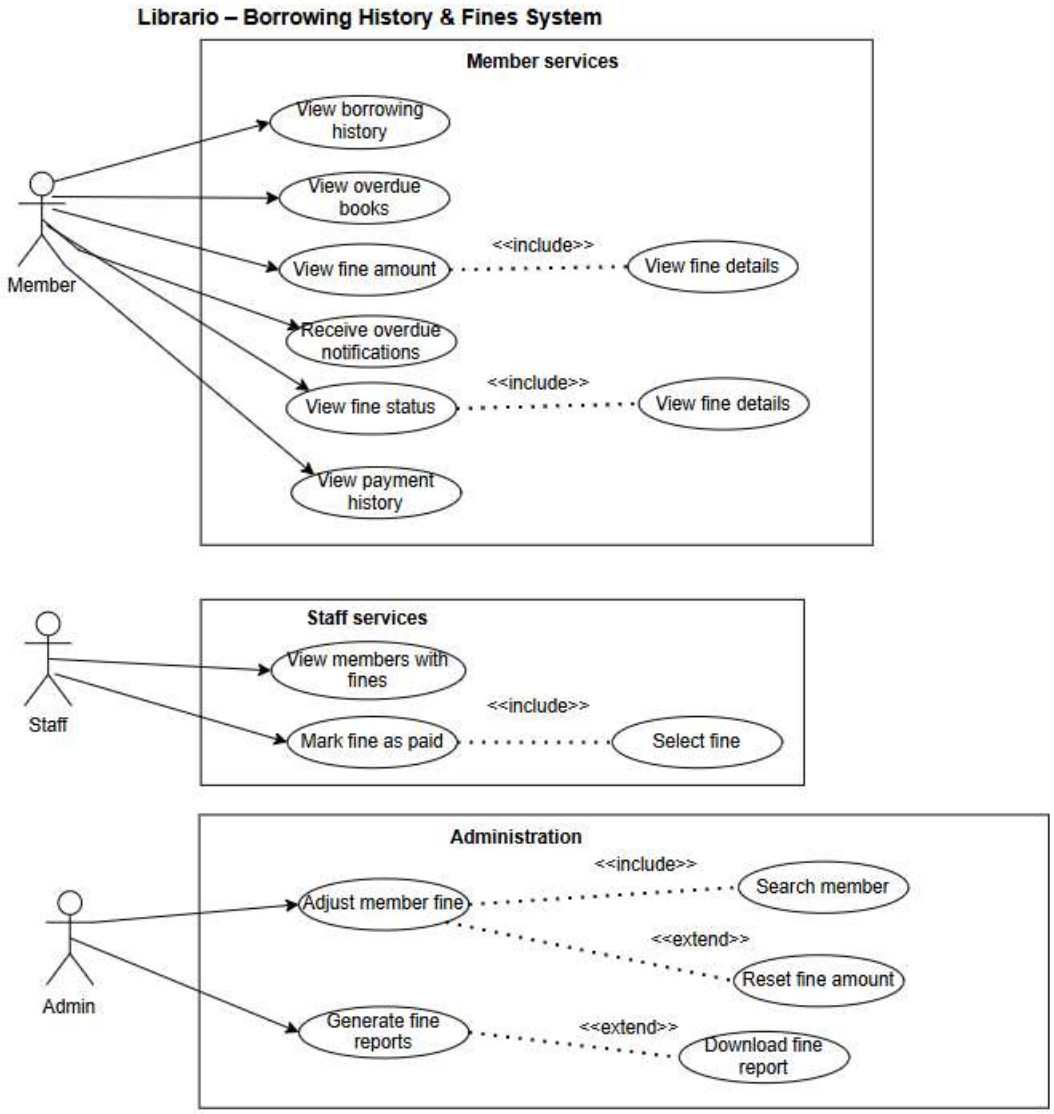
UC_09: View payment history

UC Name	UC_09 — View payment history
Dependency	None
Actors	Primary actor: Member
Preconditions	Member is logged into the system. At least one fine has been paid.
Description of the Main Sequence	1. Member navigates to the payment history section. 2. System retrieves all paid fine records for the member. 3. System displays the payment history with details such as amount, payment date, and related book.
Description of the Alternative Sequence	• No payment history: System displays **"No payment history found."** • Data retrieval error: System shows an error message.
Non-functional requirements	Payment history must load within 2 seconds. Data must be accurate and clearly presented.
Postconditions	Member successfully views their payment history.

UC_10: Generate fine reports

UC Name	UC_10 — Generate fine reports
Dependency	None
Actors	Primary actor: Administrator
Preconditions	Administrator is logged into the system. Fine records exist in the system.
Description of the Main Sequence	1. Administrator navigates to the reports section. 2. Administrator selects a report type related to fines or overdue penalties. 3. System collects the required data. 4. System generates and displays the report. 5. Administrator can review or download the report.
Description of the Alternative Sequence	• No data available: System displays **"No report data available."** • Report generation error: System shows an error message.
Non-functional requirements	Report generation must complete within a reasonable time. Report data must be accurate and clearly presented.
Postconditions	Administrator successfully views or downloads the fine report.

Use Case Diagrams



Functional Requirements

Req#	Requirement	Comments	Priority	Date	Reviewed / Approved
FR_01	The system shall allow members to view their borrowing history.	Must display book title, borrow date, return date, and status.	1	02/04/2026	Meriklesta Hasanaj
FR_02	The system shall identify and display overdue books for a member.	Overdue status is based on due date compared to current date.	1	02/04/2026	Meriklesta Hasanaj
FR_03	The system shall calculate and display the total fine amount for overdue books.	Fine calculation must be accurate and updated in real time.	1	02/04/2026	Meriklesta Hasanaj
FR_04	The system shall send notifications when a borrowed book becomes overdue.	Notification may be system-based or via email.	2	02/04/2026	Meriklesta Hasanaj
FR_05	The system shall display whether fines are paid or unpaid.	Status must update immediately after payment.	1	02/04/2026	Meriklesta Hasanaj
FR_06	The system shall allow librarians to view members with unpaid or overdue fines.	Must include member details and fine amounts.	2	02/04/2026	Meriklesta Hasanaj
FR_07	The system shall allow administrators to adjust or reset member fines.	Changes must be saved immediately and logged.	1	02/04/2026	Meriklesta Hasanaj
FR_08	The system shall allow librarians to mark fines as paid.	Fine status must update instantly in the system.	1	02/04/2026	Meriklesta Hasanaj
FR_09	The system shall allow members to view their payment history.	Must include payment amount, date, and related book.	2	02/04/2026	Meriklesta Hasanaj
FR_10	The system shall allow administrators to generate reports about	Reports must include collected fines and outstanding penalties.	2	02/04/2026	Meriklesta Hasanaj

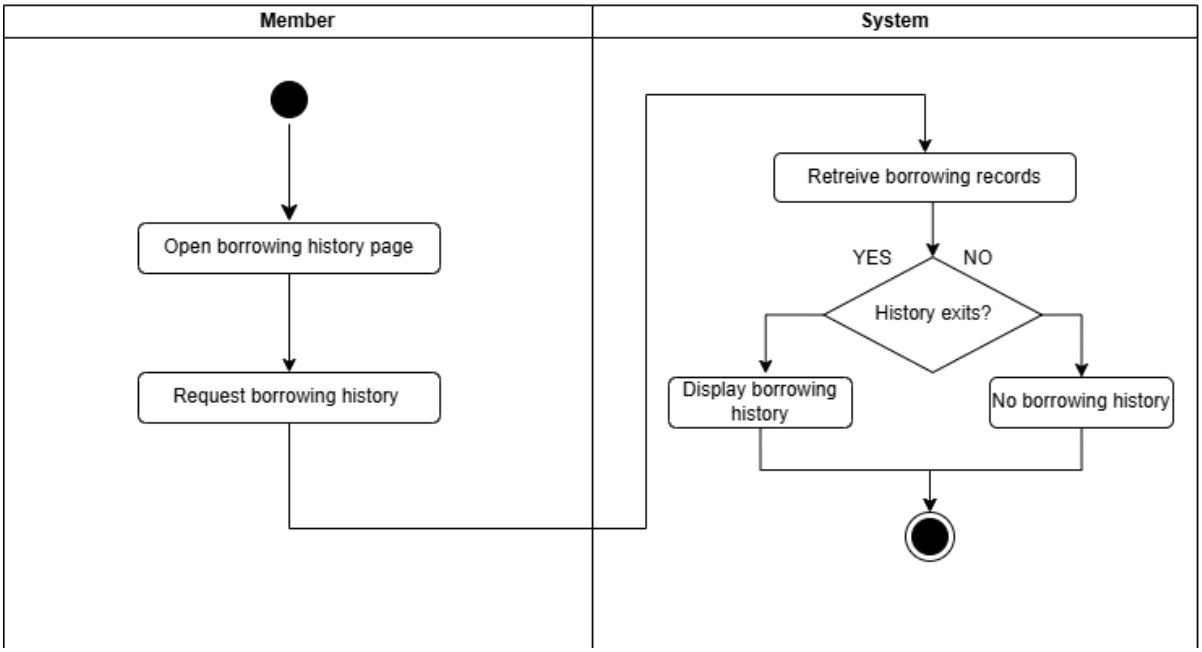
	fines and overdue penalties.				

Non-Functional Requirements

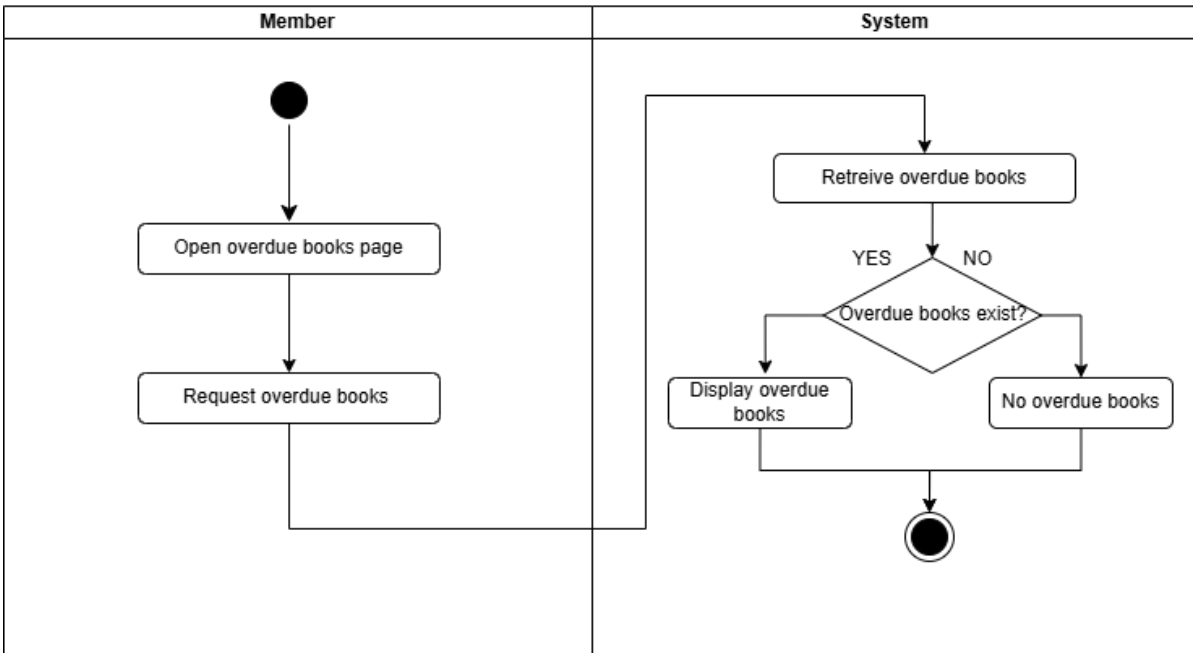
Req#	Category	Requirement
NFR_01	Performance	The system must display borrowing history, overdue books, and fines within 2 seconds under normal load.
NFR_02	Security	Only authorized users (members, staff, administrators) must access their respective functionalities.
NFR_03	Security	Fine data and payment information must be protected from unauthorized access or modification
NFR_04	Reliability	The system must ensure accurate calculation of fines at all times.
NFR_05	Usability	All lists (borrowing history, overdue books, fines) must be displayed in a clear and user-friendly format.
NFR_06	Usability	Error messages must be clear and understandable without exposing system details.
NFR_07	Reliability	Notifications for overdue books must be delivered reliably without significant delay.
NFR_08	Scalability	The system must handle multiple users accessing borrowing and fine information simultaneously without performance issues.
NFR_09	Compliance	User data related to borrowing and fines must comply with data protection regulations.
NFR_10	Auditability	All actions related to fines (payments, adjustments) must be logged with timestamp and user/admin identification.

ACTIVITY DIAGRAM

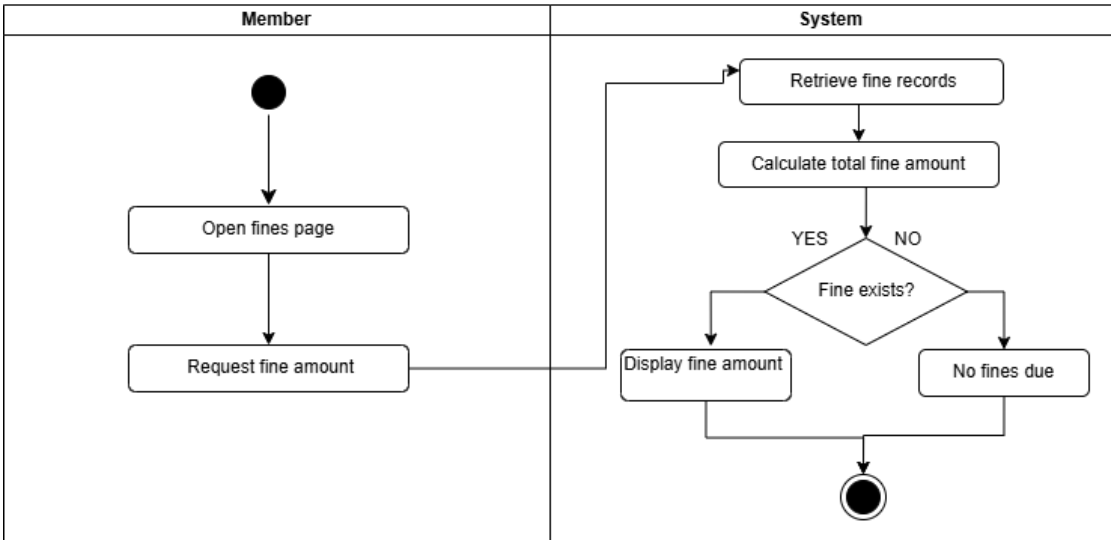
Activity Diagram: View Borrowing History



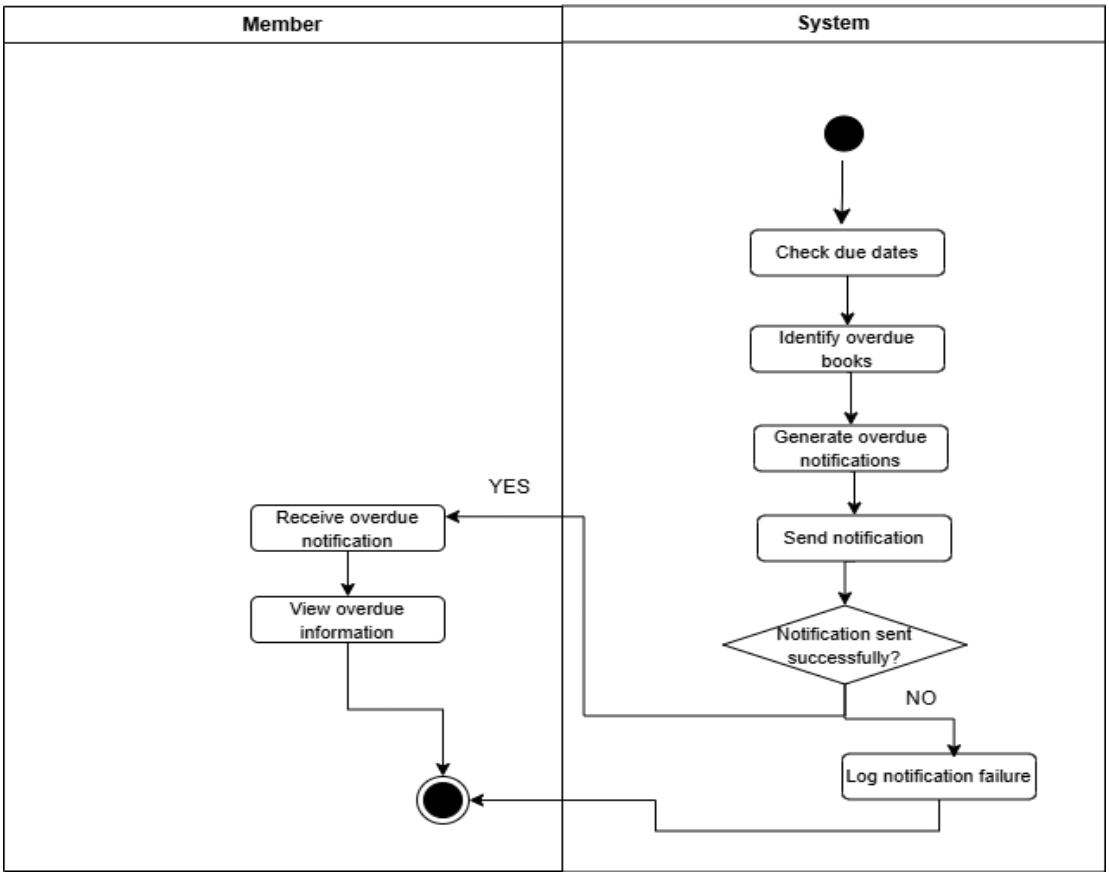
Activity Diagram : View Overdue Books



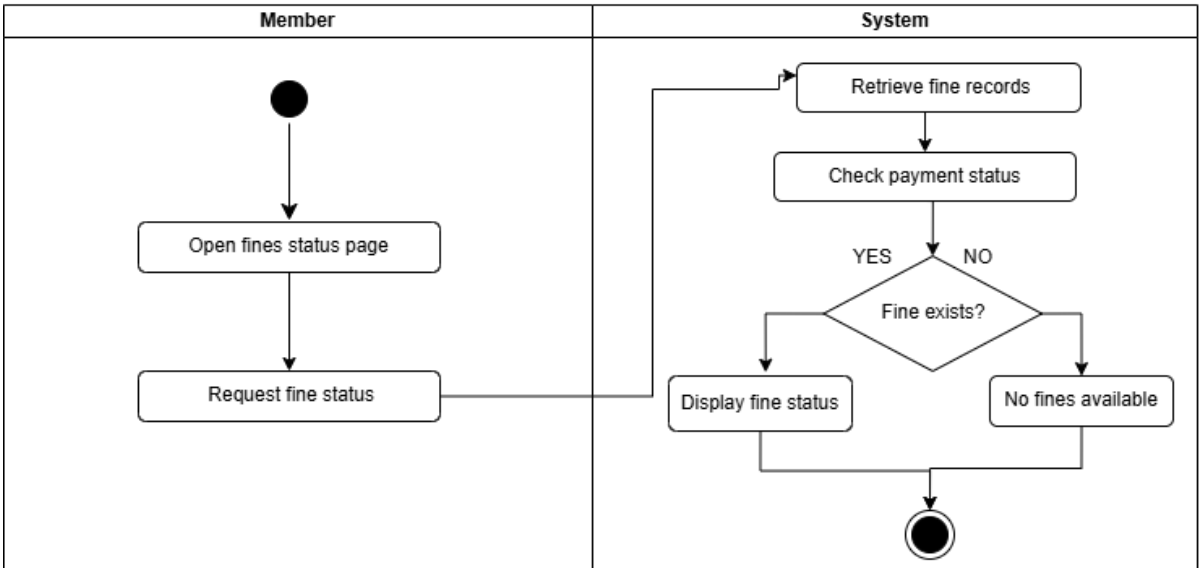
Activity Diagram: View fine amount



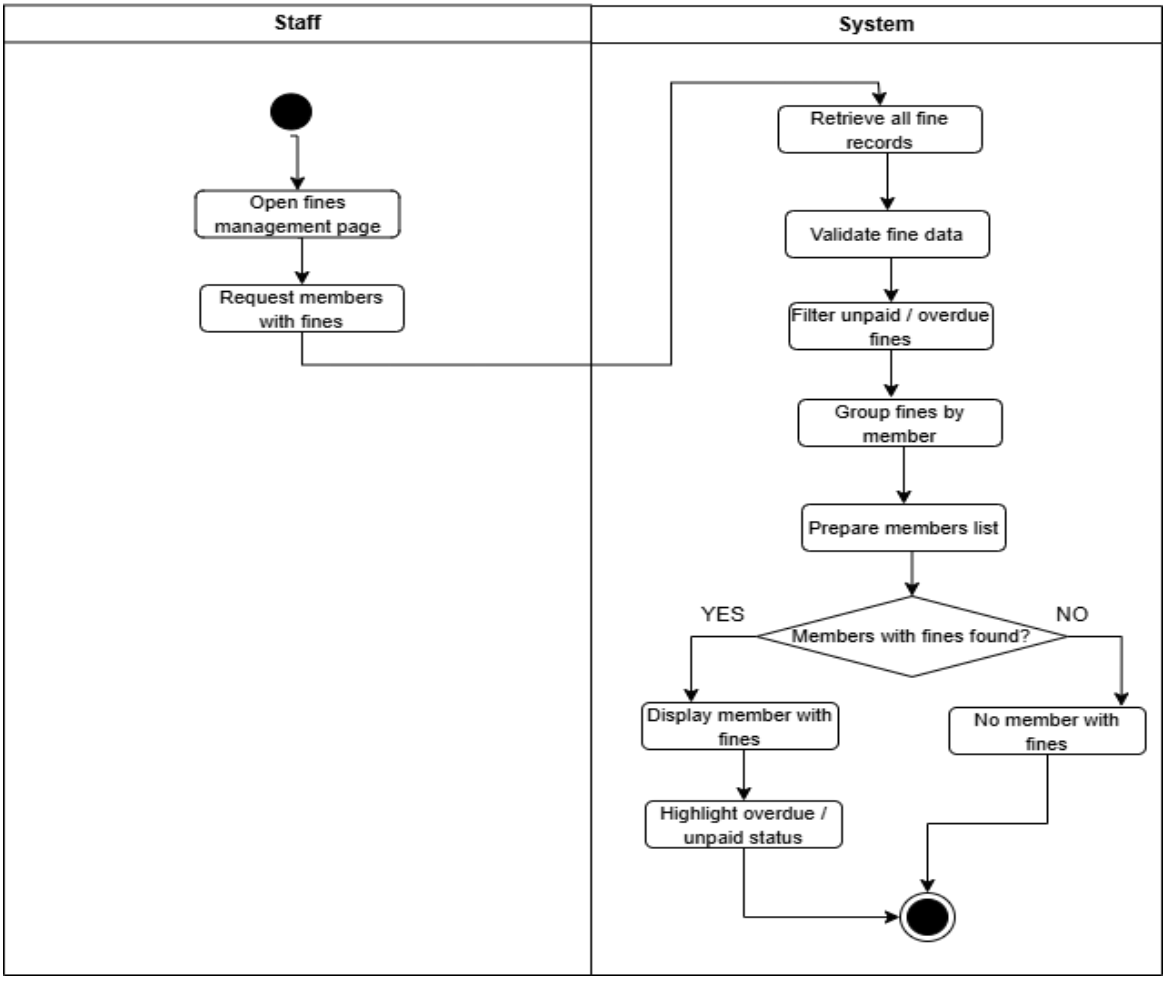
Activity Diagram: Receive overdue notifications



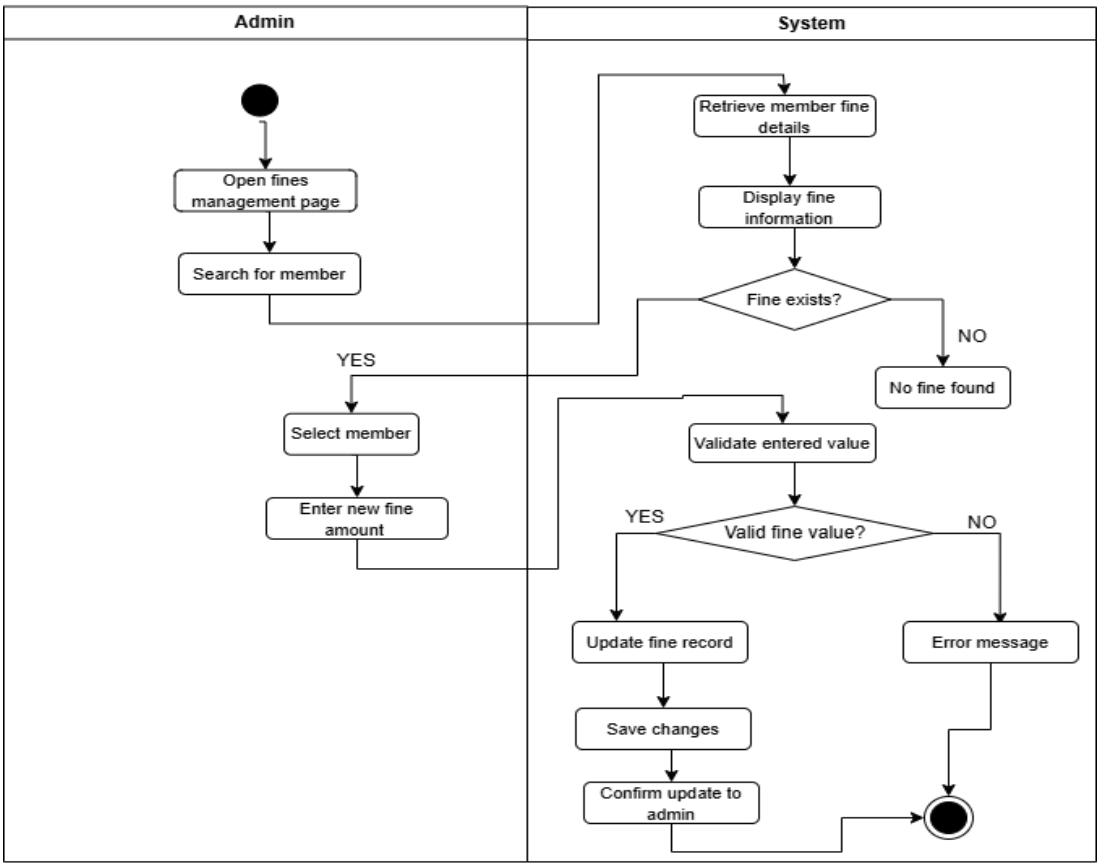
Activity Diagram: View Fine Status



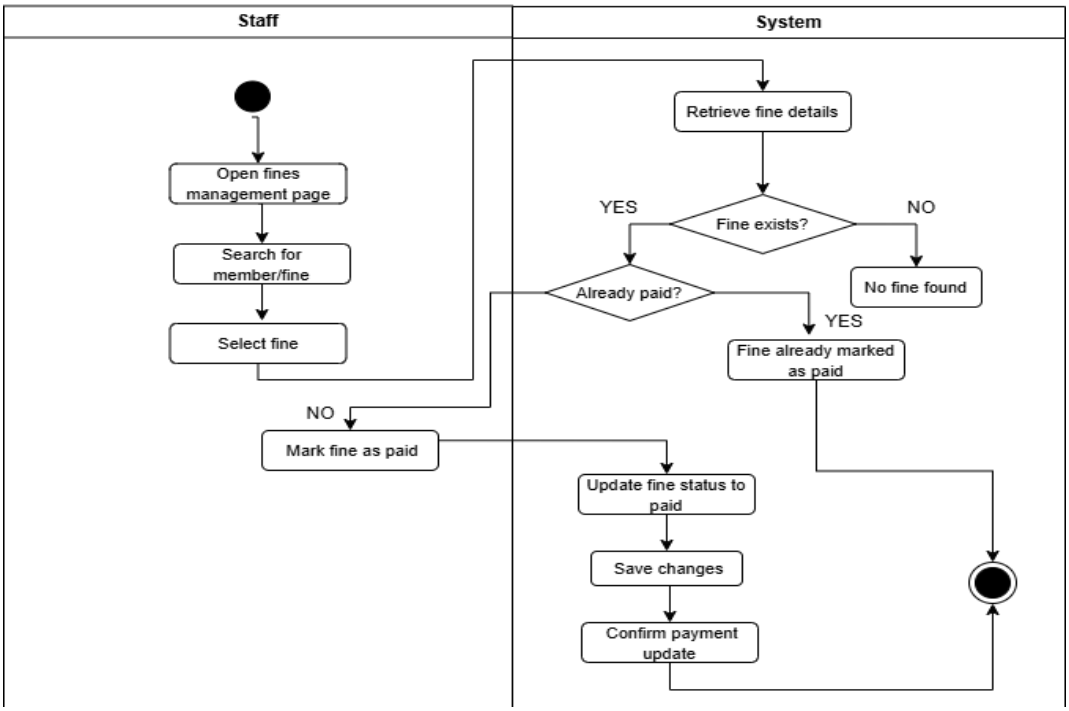
Activity Diagram: View members with fines



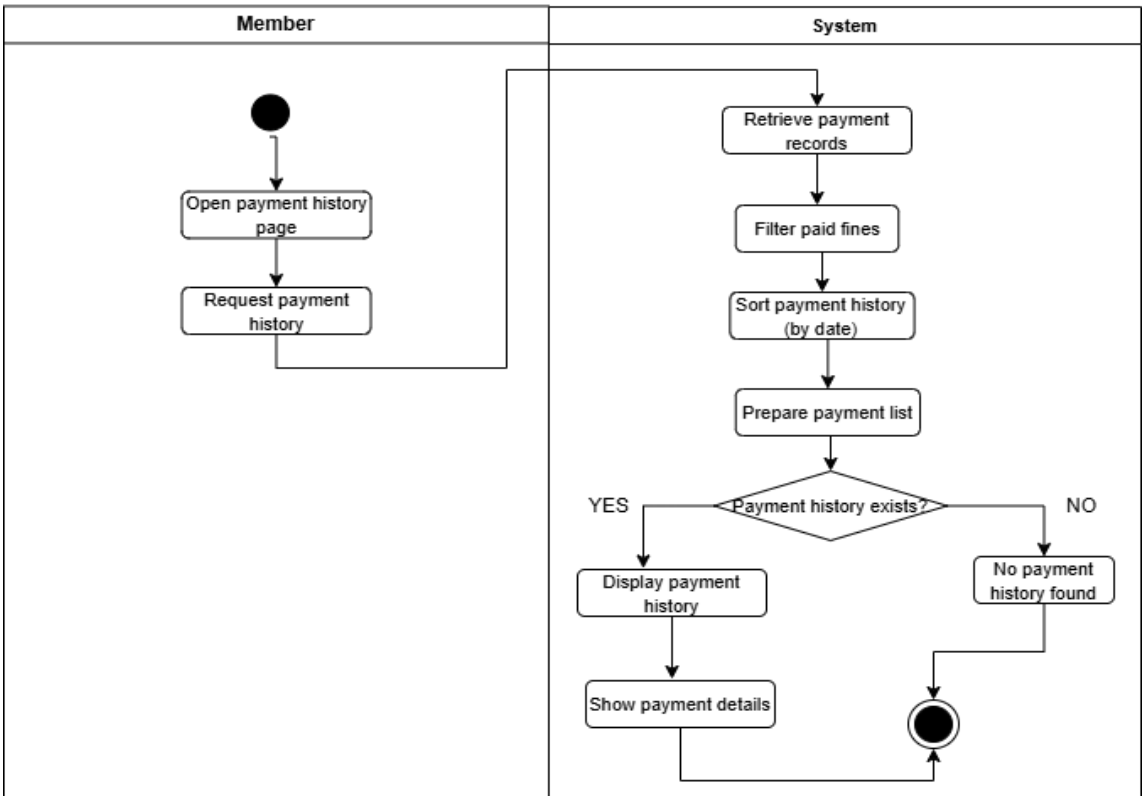
Activity Diagram: Adjust member fine



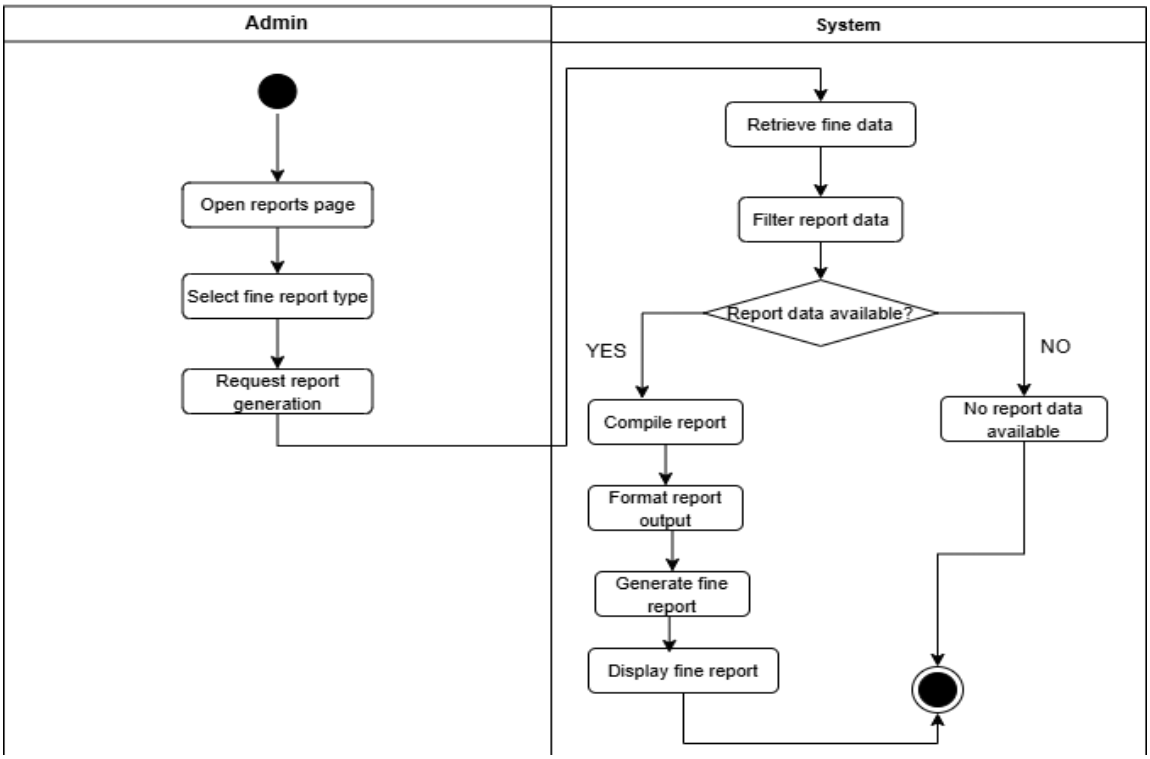
Activity Diagram: Mark fine as paid



Activity Diagram: View payment history

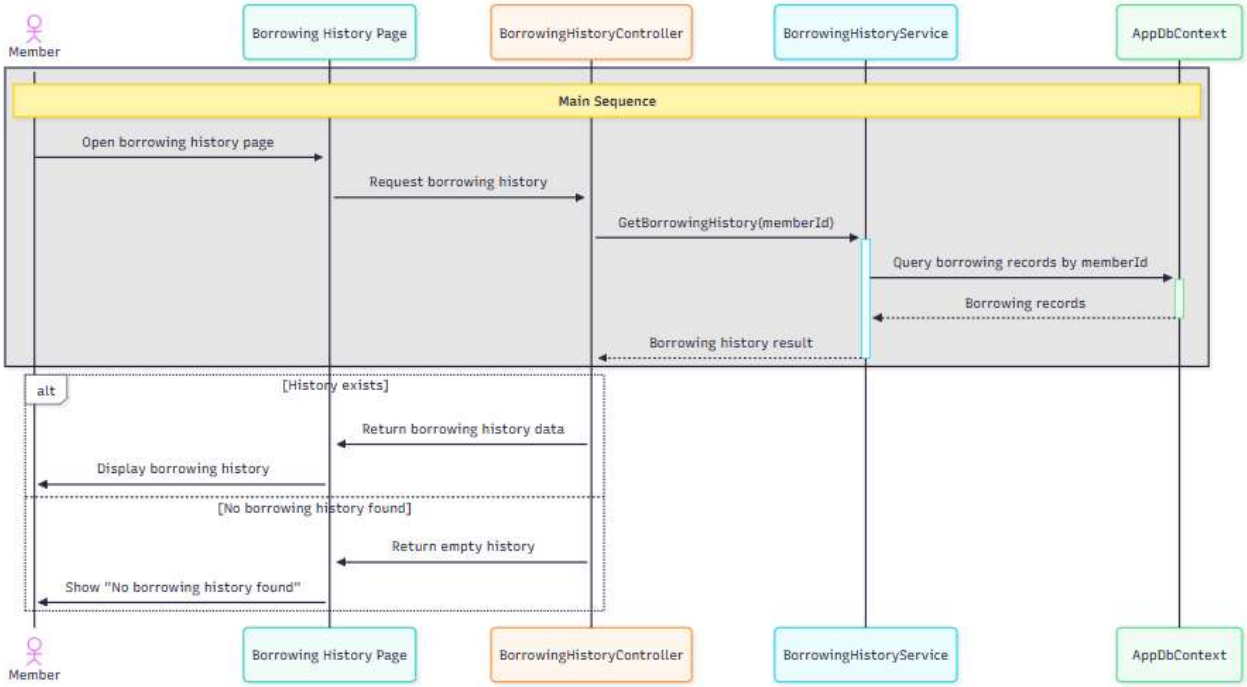


Activity Diagram: Generate fine reports

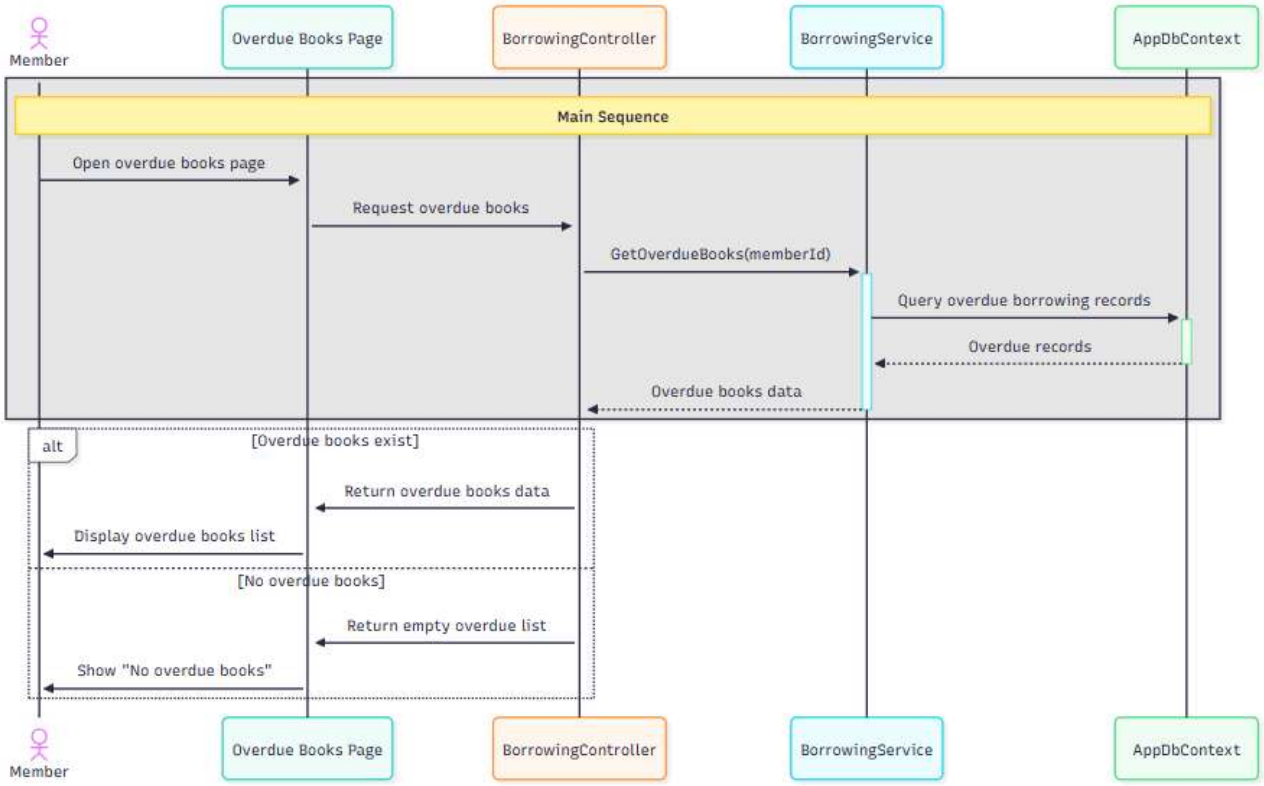


SEQUENCE DIAGRAM

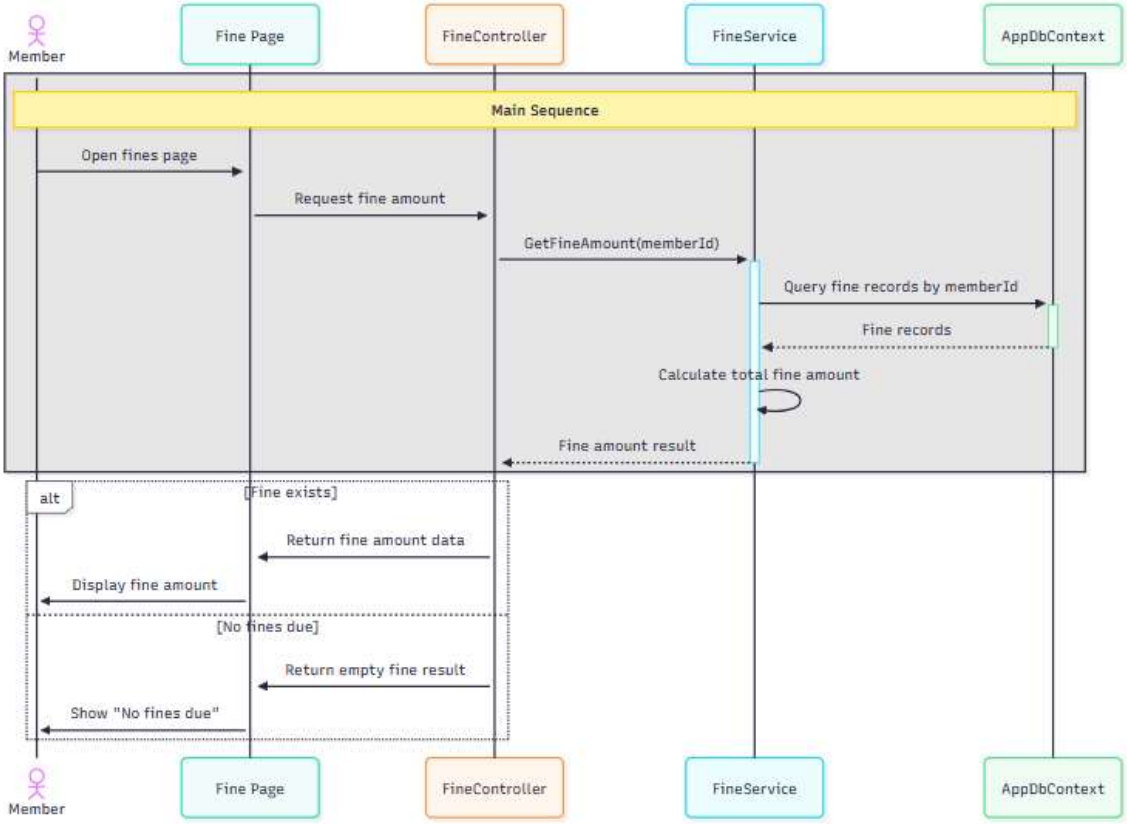
Sequence Diagram-->View Borrowing History



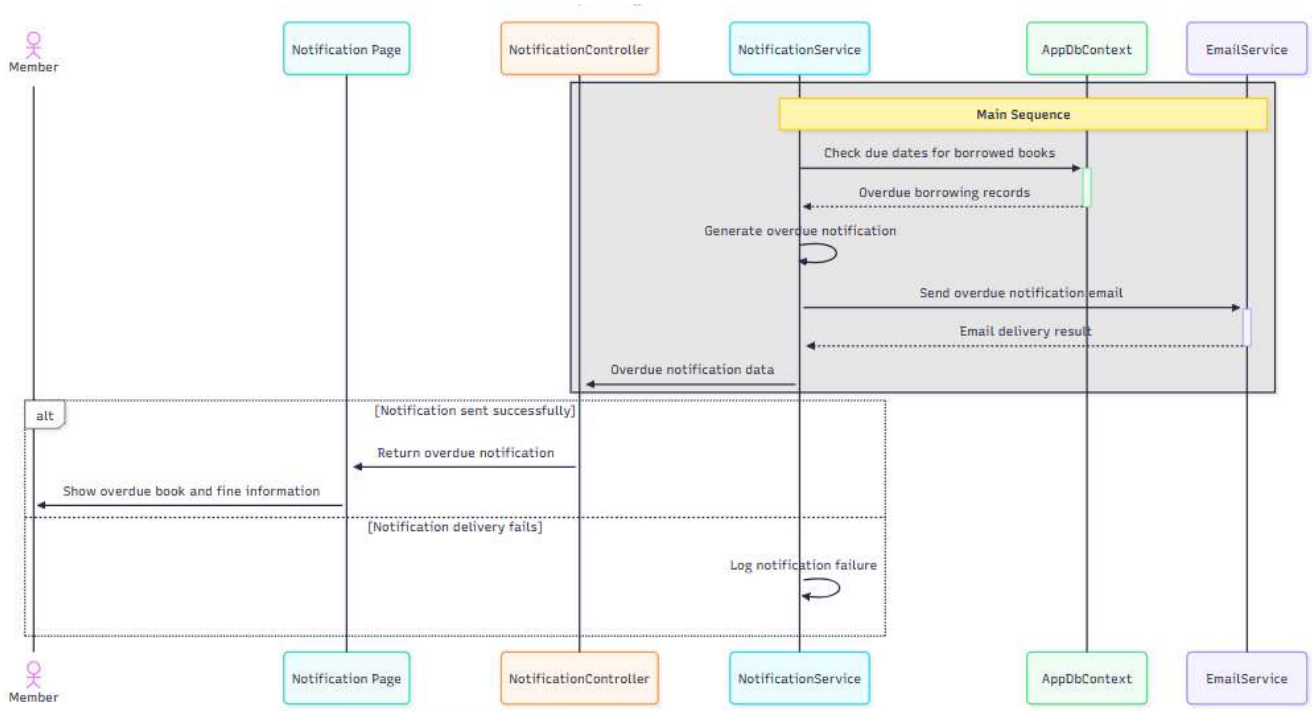
Sequence Diagram-->View Overdue Books



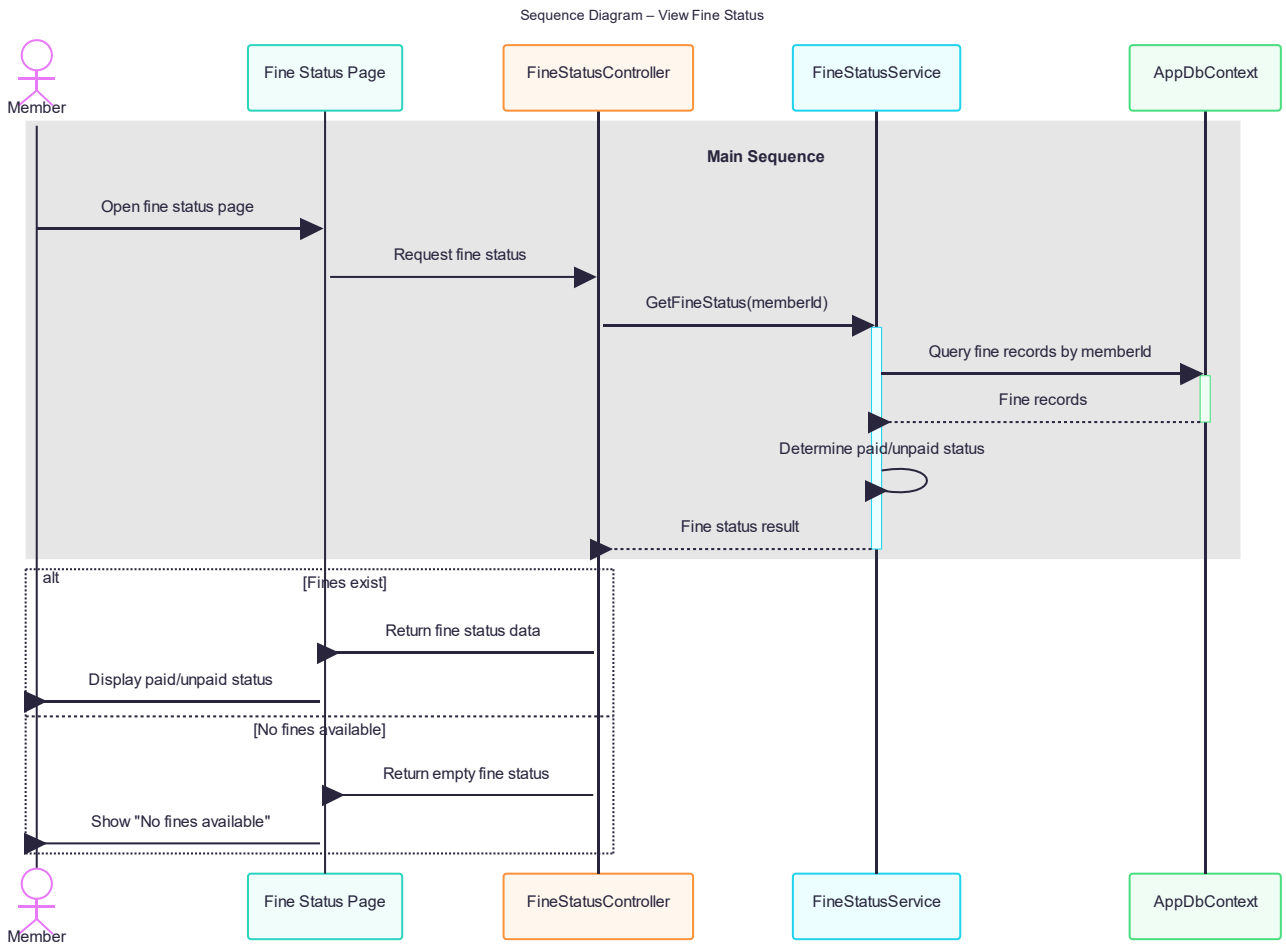
Sequence Diagram-->View Fine Amount



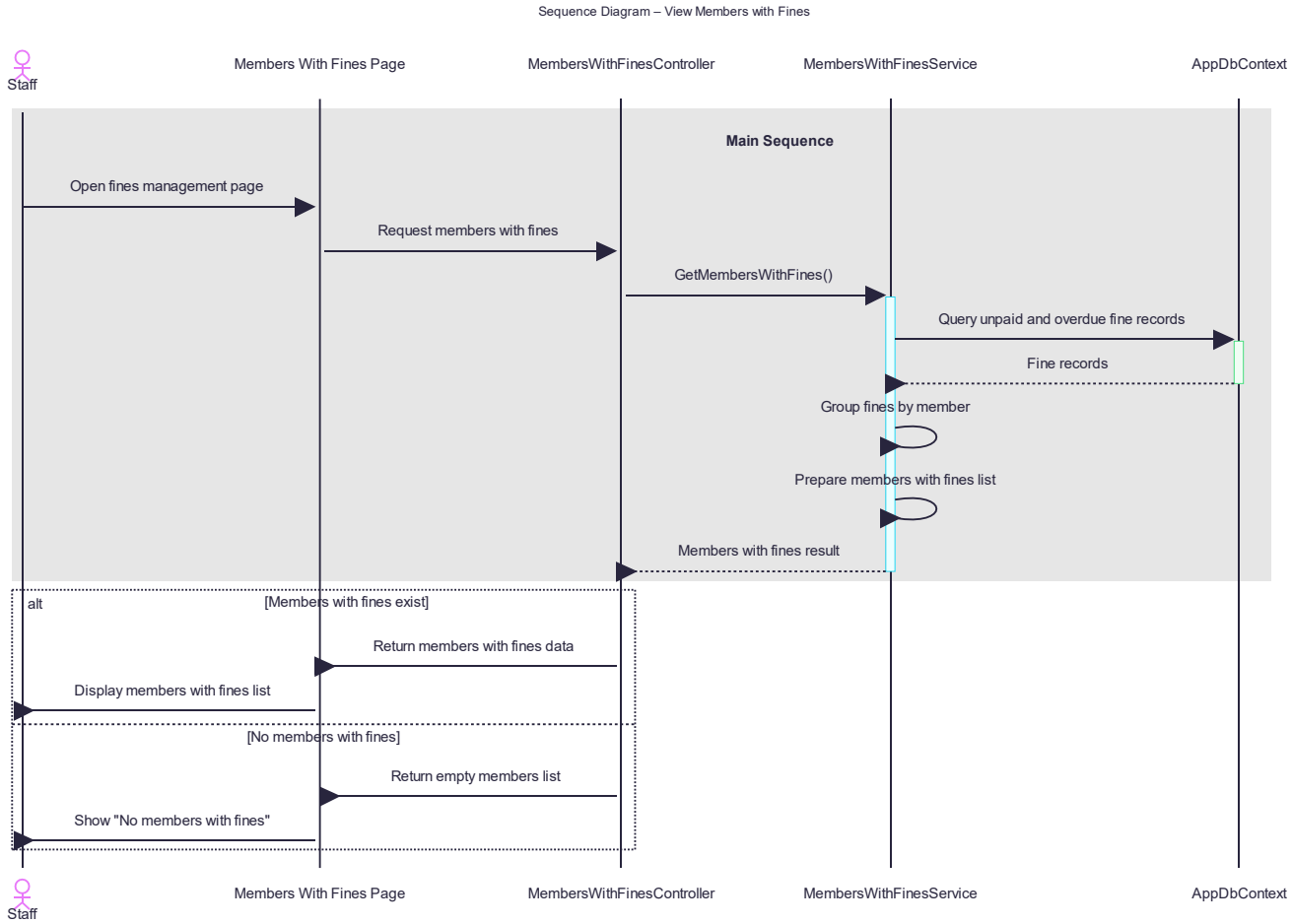
Sequence Diagram-->Receive Overdue Notifications



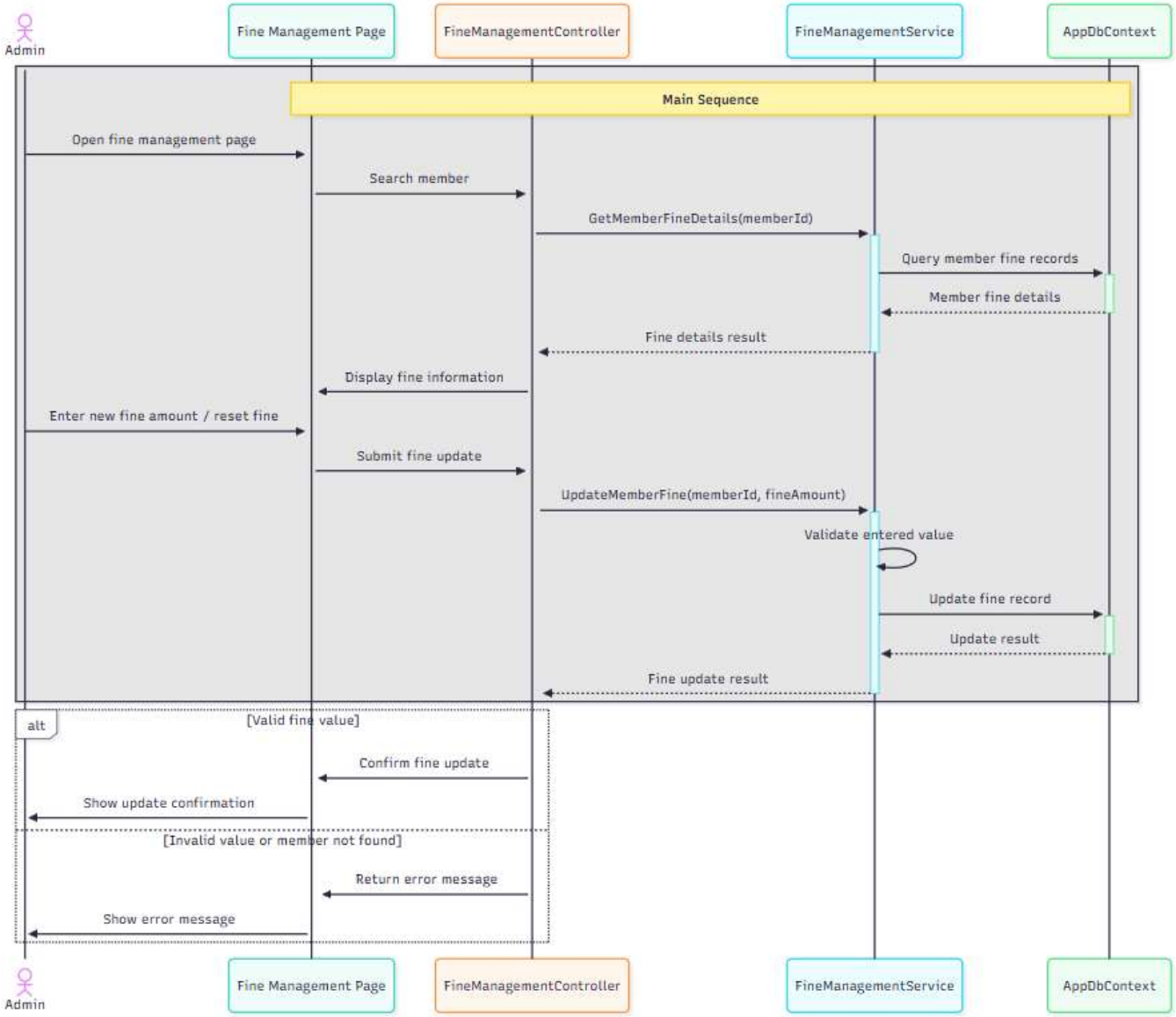
Sequence Diagram-->View Fine Status



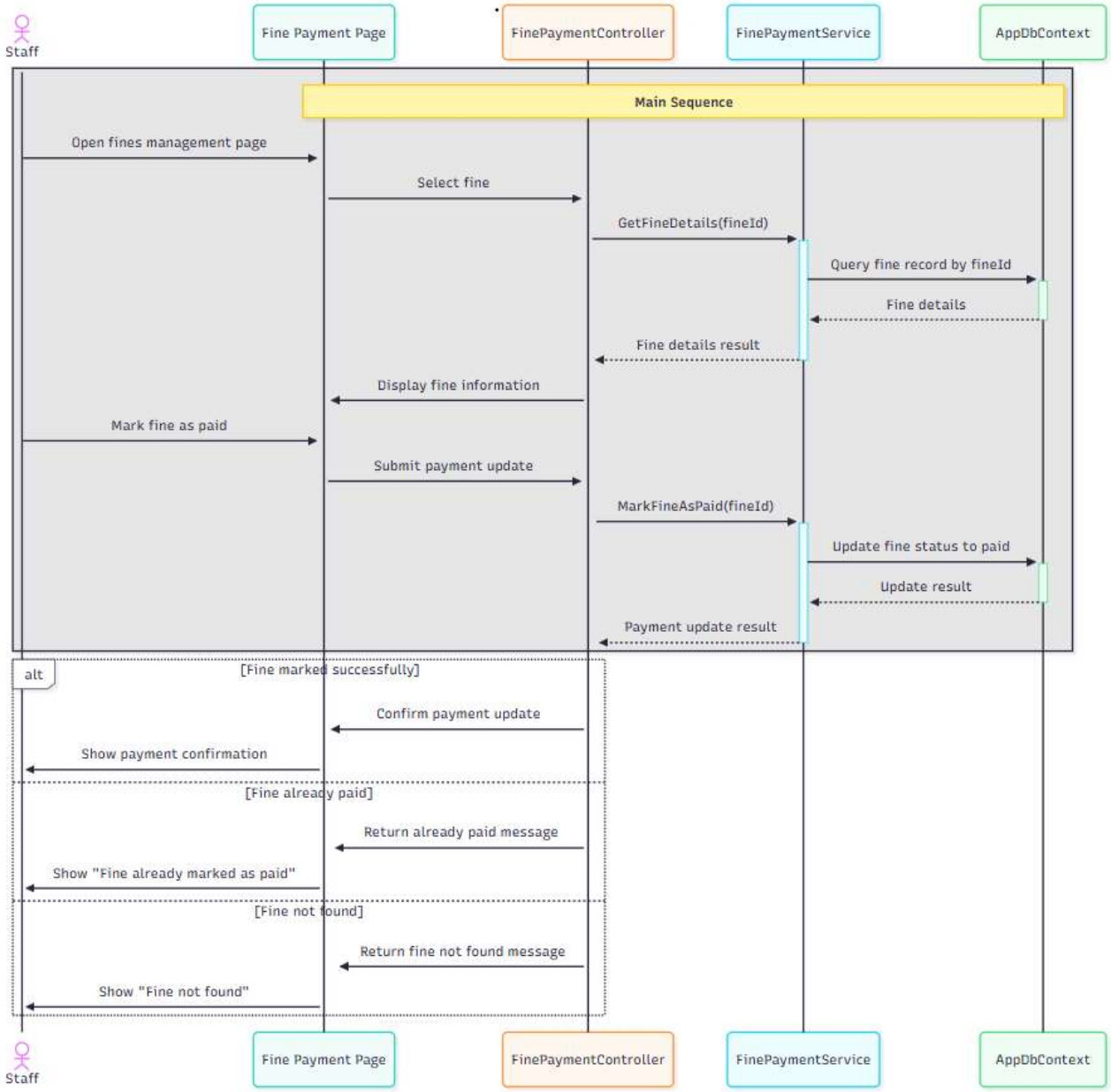
Sequence Diagram-->View members with fines



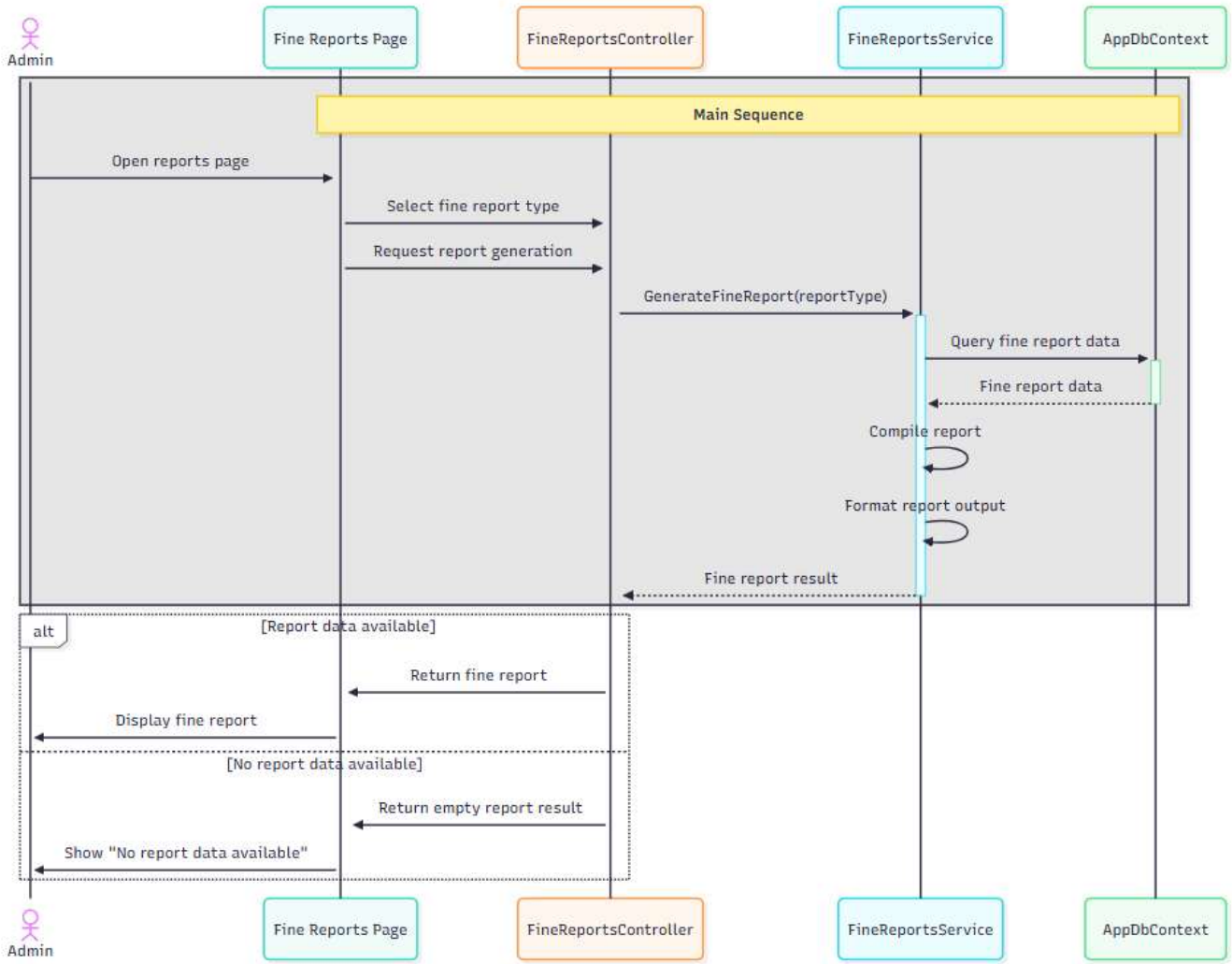
Sequence Diagram-->Adjust member fine



Sequence Diagram-->Mark fine as paid

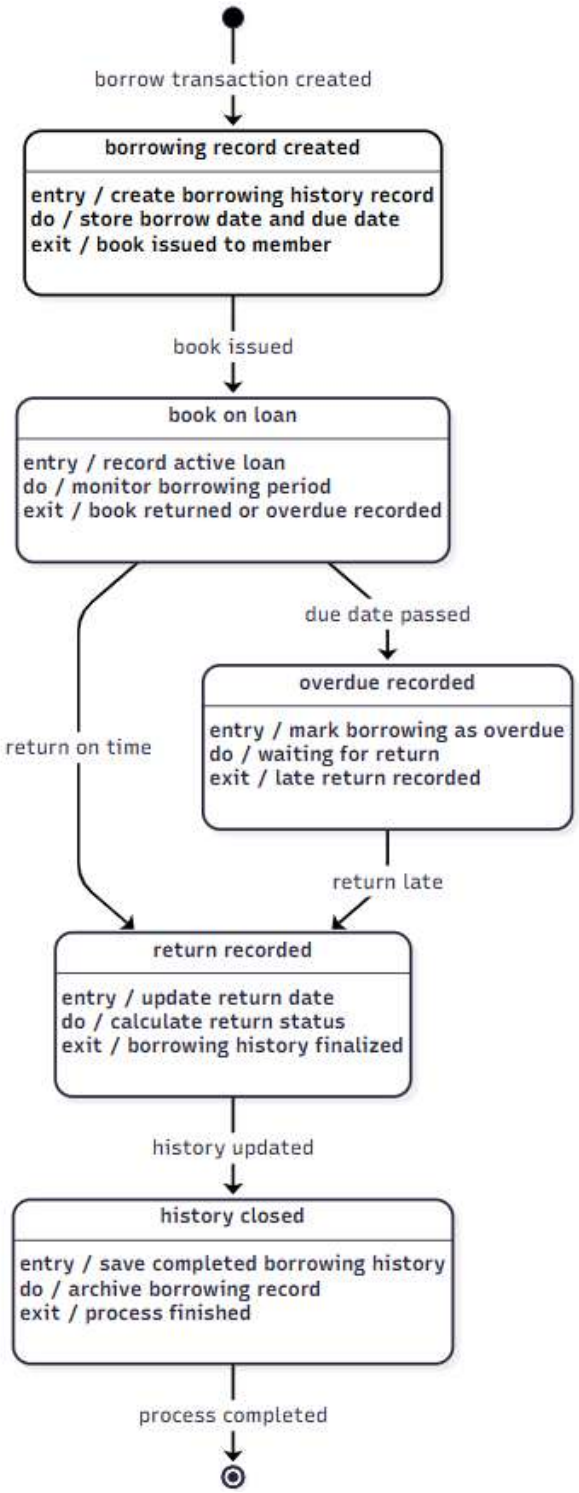


Sequence Diagram-->Generate fine reports

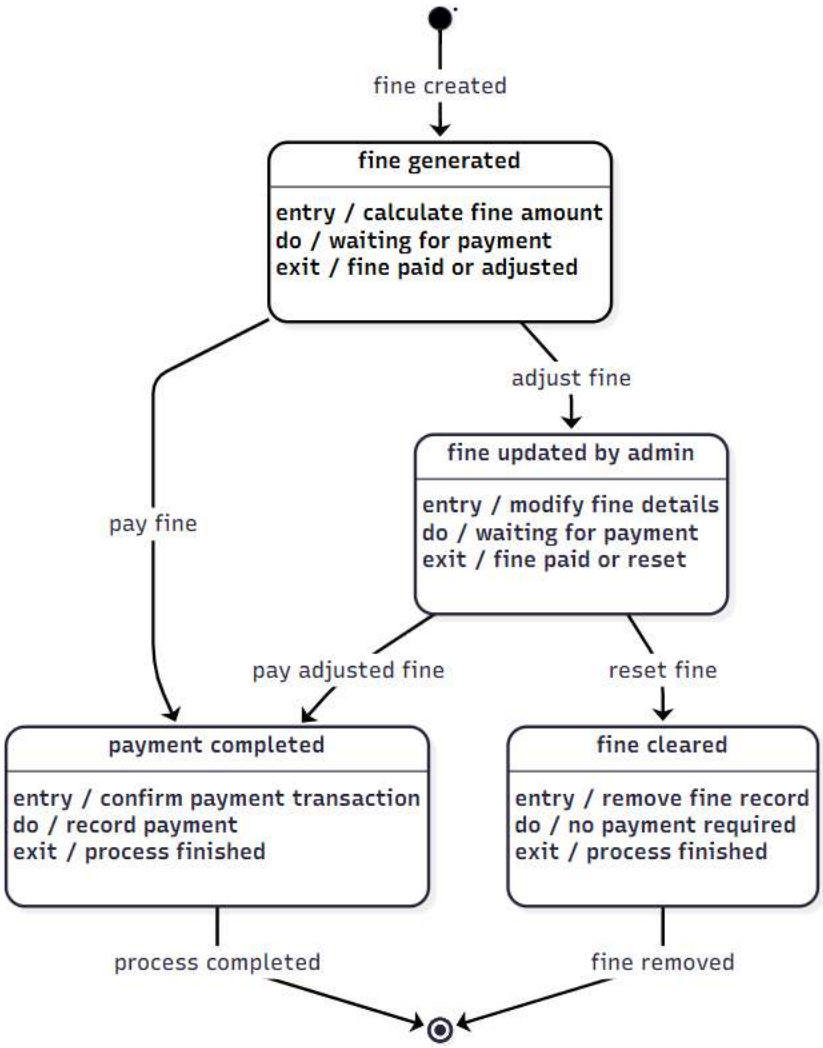
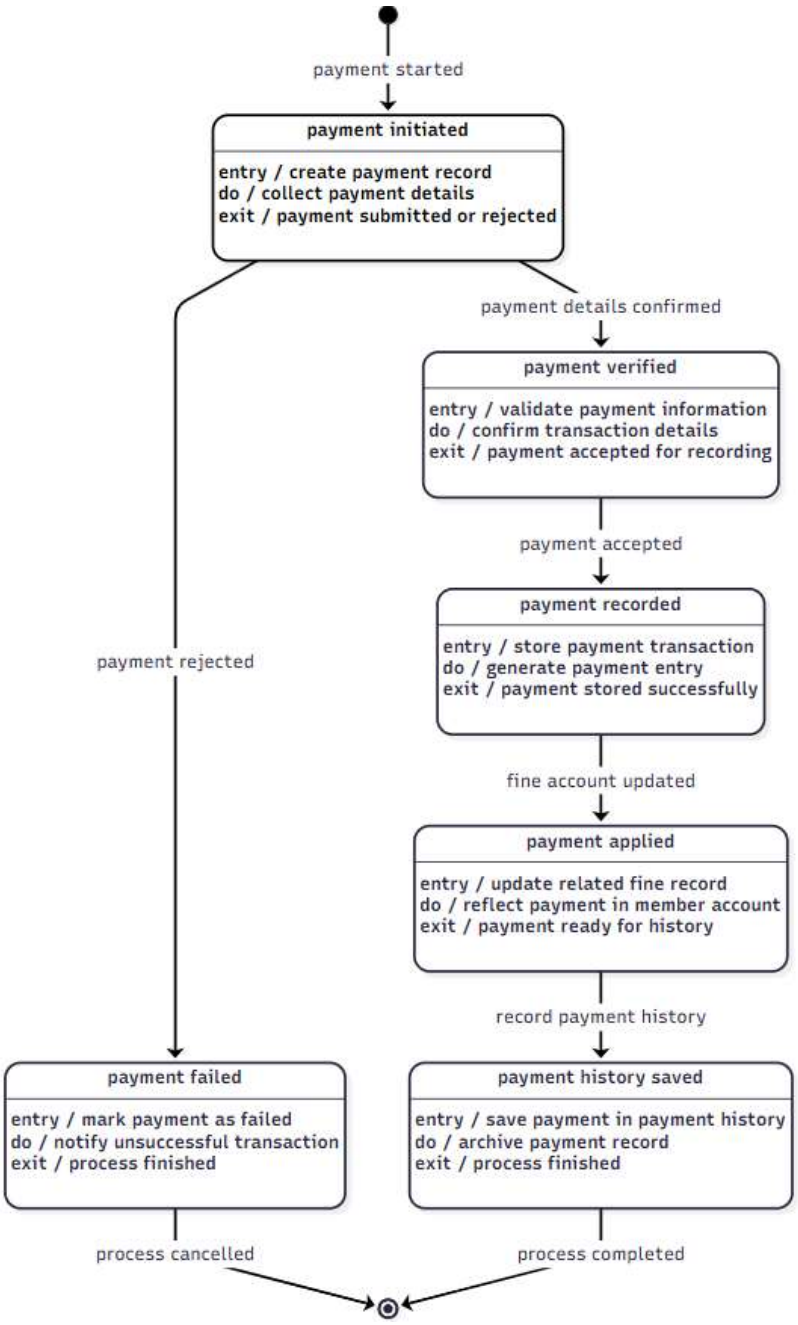


STATE DIAGRAM

State Diagram – Borrowing History Lifecycle



State Diagram – Fine Management Lifecycle



State Diagram – Payment Record Lifecycle